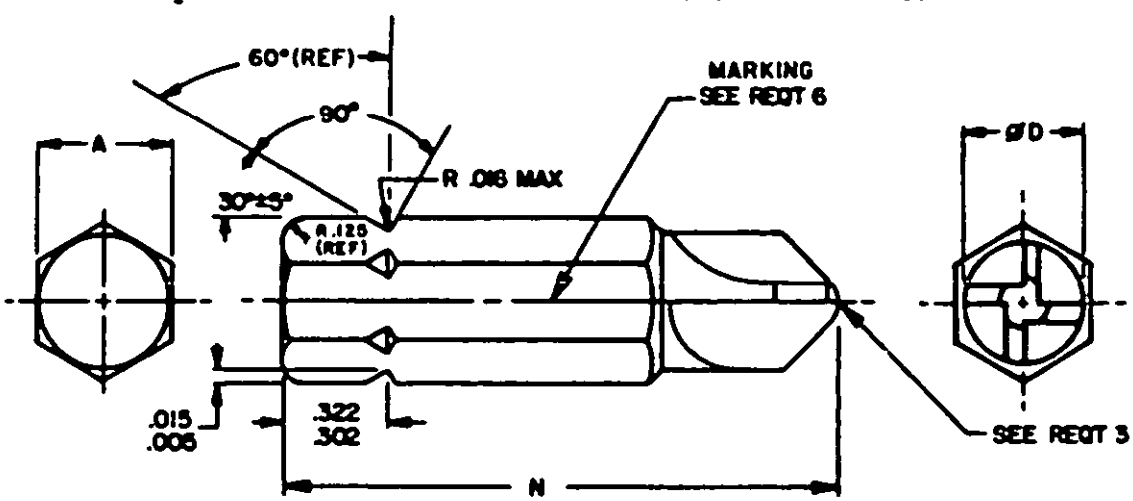
MILITARY SPECIFICATION SHEET

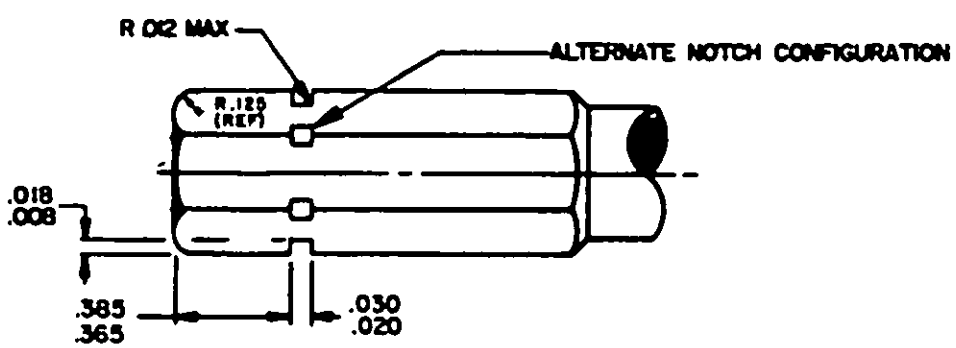
BIT, SCREWDRIVER, TORQ-SET, EXTERNAL HEX SHANK, MANUALLY DRIVEN

NON-RIBBED DRIVERS WITH DASH NUMBERS 6, 8, 8A, 10, 10A, 250A AND 312A ARE NOT RECOMMENDED FOR DOD USE, SEE MIL-B-9946/16 FOR RIBBED DRIVERS IN THESE SIZES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the bits described herein shall consist of this specification and the latest issue of MIL-B-9946.





NOTES:
1. All dimensions are in inches.
2. Bit and retaining groove shall be in accordance with TABLE I.

FIGURE 1. BIT, SCREWDRIVER, TORQ-SET, EXTERNAL HEX SHANK, MANUALLY DRIVEN

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TABLE I. DASH NUMBERS AND DIMENSIONS

DASH NUMBER	DRIVER SIZE	RECESS SIZE	A HEX +.000 -.004	Ø D SHANK	N ±.120 BIT LENGTH	ULTIMATE TORQUE MIN LBf - IN	FATIGUE TORQUE LBf - IN
0	# 0	# 0	.250	.185 / .180	1.00	—	—
1	# 1	# 1	.250	.185 / .180	1.00	—	—
2	# 2	# 2	.250	.185 / .180	1.00	—	—
3	# 3	# 3	.250	.185 / .180	1.00	—	—
4	# 4	# 4	.250	.185 / .180	1.00	—	—
5	# 5	# 5	.250	.185 / .180	1.00	25	12
6	# 6	# 6	.250	.185 / .180	1.00	35	16
8	# 8	# 8	.250	.247 / .242	1.00	65	30
8A	# 8	# 8	.312	.309 / .304	1.25		
10	#10	#10	.250	.247 / .242	1.00	100	45
10A	#10	#10	.312	.309 / .304	1.25		
250A	1/4	1/4	.312	.309 / .304	1.25	250	115
312A	5/16	5/16	.312	.309 / .304	1.25	500	200
375	3/8	3/8	.437	.434 / .429	1.25	675	303
437	7/16	7/16	.437	.434 / .429	1.25	1,375	619
500	1/2	1/2	.625	.622 / .617	1.25	2,125	957
562	9/16	9/16	.625	.622 / .617	1.25	3,000	1,350
625	5/8	5/8	.625	.622 / .617	1.25	4,050	1,823

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REQUIREMENTS:

1. Material: Alloy Tool Steel in accordance with American Society for Testing Material (ASTM) A681, Shock-Resisting Steel, Type S-2 (UNS T41902).

2. Hardness: Macrohardness, (Rockwell C Scale) 56-62HRC. Three macrohardness readings shall be taken at the locations shown in FIGURE 2. Each hardness value must be within the range specified.

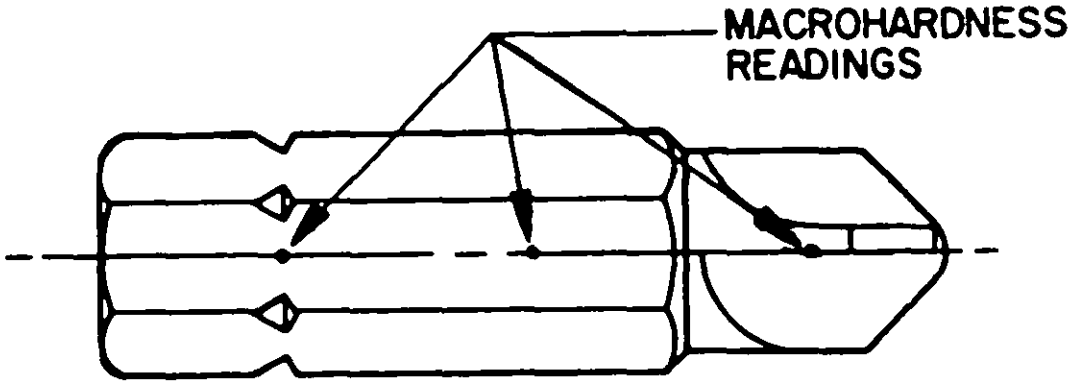


FIGURE 2. MACROHARDNESS READING LOCATIONS

Superficial hardness, (Rockwell HR15N Scale) 87-90. Superficial hardness readings shall be taken on bid samples at two locations on sectionalized specimens along the blade.

3. Tip point: Tip point shall be in accordance with MS14191.

4. Driver bit gage: Measures the alignments of the opposed wing surfaces and the maximum and minimum material conditions of the driver by means of a "GO" and "NO GO" criteria. The driver tip profile notch on this gage measures, by eye, the dimensional accuracy of the driver tip profile. The driver bit gage shall be in accordance with FIGURE 3 and TABLE II.

5. Test block: As shown in FIGURE 4 is used to conduct torque tests on the tool bit. This test block shall have the recess penetration values, as specified in TABLE III, in order to properly conduct the required torque tests as described herein and in MIL-B-9946.

6. Identification marking: Driver bits shall be marked by etching, stamping or engraving onto each bit the following minimum information:

   a. Military specification part number.

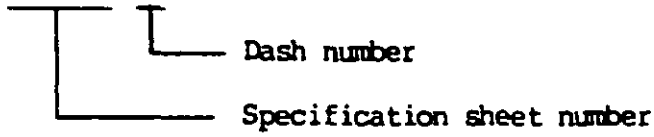
   b. Manufacturer's symbol or code.

   c. Country of manufacture.

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7. Part number: The part number consists of the letter "M", the basic number of this specification sheet, followed by a dash number taken from TABLE I.

Example: M9946/4-8A



M9946/4-8A indicates - Bit, Screwdriver, Torq-Set, External Hex Shank, Manually Driven; Driver Size #8; Recess Size #8; Hex .312; Length 1.25.

8. Torq-Set® is a registered trademark of Phillips Screw Company. The Government does not have a royalty-free license.

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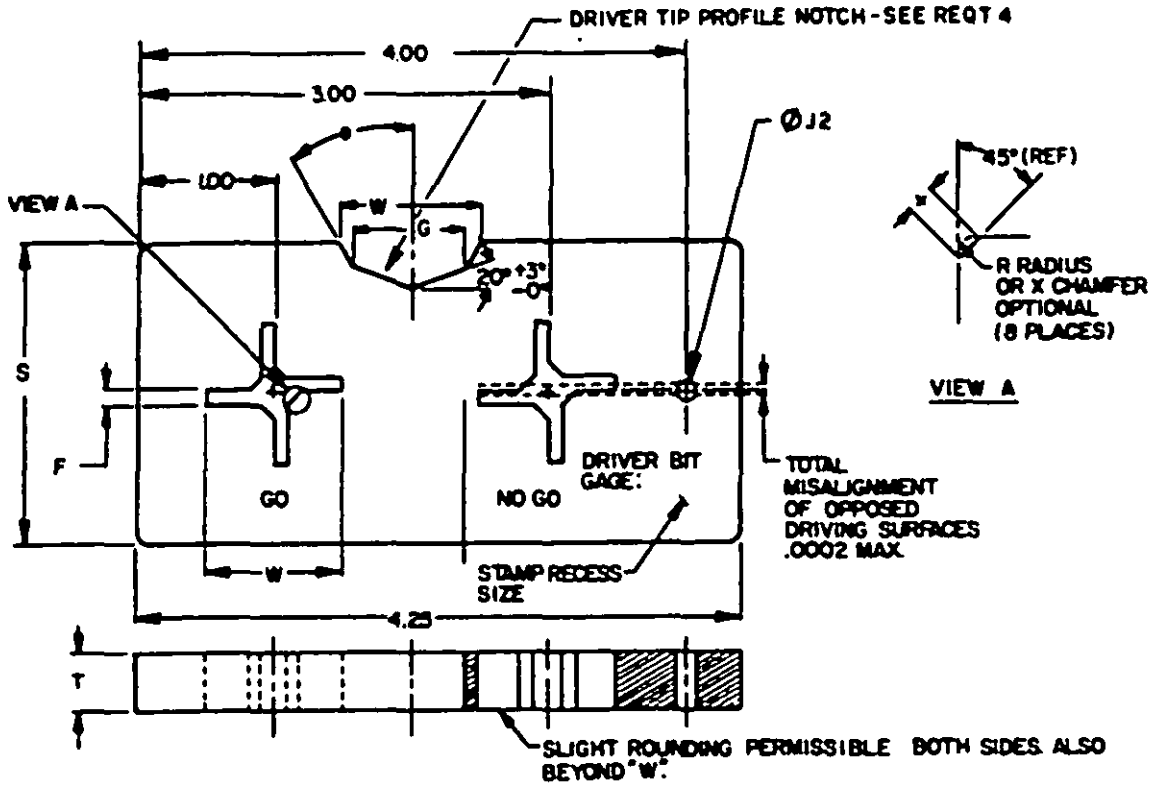


FIGURE 3. DRIVER BIT GAGE

TABLE II. DIMENSIONS FOR OFFSET CRUCIFORM DRIVER BIT GAGE

RECESS SIZE NOMINAL	S MIN WIDTH	T ±.005 THICK	GO GAGE DIMENSIONS						NO-GO GAGE DIMENSIONS					
			W ±.005	G ±.005	F +.0002 -.0000	R +.000 -.001	X +.000 -.001	θ +1° -0°	W ±.005	G ±.005	F +.0000 -.0002	R +.000 -.001	X +.000 -.001	θ +1° -0°
# 0	1.00	.187	.082	.038	.0155	.007	.010	47	SAME AS GO DIMENSIONS	SAME AS GO DIMENSIONS	.0140	SAME AS GO DIMENSIONS	SAME AS GO DIMENSIONS	SAME AS GO DIMENSIONS
# 1	1.00	.187	.099	.045	.0155	.007	.010	47			.0140			
# 2	1.00	.187	.117	.053	.0155	.007	.010	47			.0140			
# 3	1.00	.187	.135	.060	.0175	.007	.010	47			.0160			
# 4	1.00	.187	.152	.068	.0185	.012	.017	47			.0170			
# 5	1.00	.187	.170	.076	.0215	.012	.017	47			.0200			
# 6	1.00	.187	.188	.083	.0245	.018	.026	47			.0230			
# 8	1.00	.187	.223	.097	.0295	.018	.026	47			.0280			
#10	1.00	.187	.258	.112	.0345	.025	.036	47			.0330			
1/4	1.00	.187	.340	.148	.0465	.025	.036	48			.0435			
5/16	1.00	.187	.372	.186	.0595	.031	.044	48			.0565			
3/8	1.00	.187	.452	.222	.0705	.031	.044	48			.0675			
7/16	1.00	.187	.493	.258	.0835	.048	.068	48			.0805			
1/2	1.25	.250	.563	.294	.0945	.062	.088	48			.0915			
9/16	1.25	.250	.633	.330	.1055	.078	.110	48			.1020			
5/8	1.25	.250	.703	.367	.1175	.078	.110	48			.1140			

NOTES:

1. All dimensions are in inches.
2. Tolerance: ±.015 unless otherwise noted.
3. Material: CRES in accordance with ASTM A276, Type 440C (UNS S44004).
4. Hardness: 58-62HRC.
5. Surface texture: 35 microinches max.

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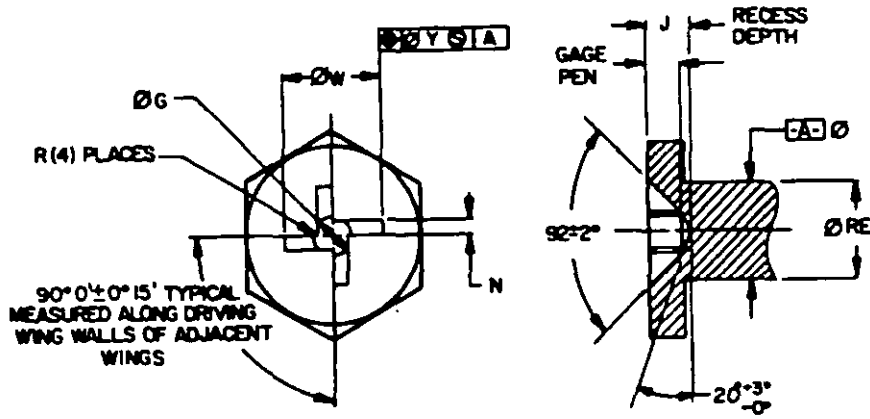


FIGURE 4. TEST BLOCK

TABLE III. DIMENSIONS FOR TEST BLOCK WITH RESTRICTED GAGE PENETRATION FOR TESTING OFFSET CRUCIFORM BITS

RECESS SIZE NOMINAL	DIMENSIONS									
	N		Ø G		R RADIUS		Ø Y TOL	Ø W WING	J RECESS DEPTH	GAGE PEN
	MAX	MIN	MAX	MIN	MAX	MIN				
# 0	.018	.016	.037	.033	.010	.000	.012	.087 .077	.033 .023	.0250 .0230
# 1	.018	.016	.044	.040				.104 .094	.039 .029	.0300 .0280
# 2	.018	.016	.052	.048				.122 .112	.046 .036	.0350 .0330
# 3	.021	.018	.059	.055				.140 .130	.053 .043	.0405 .0385
# 4	.023	.020	.067	.063	.018	.008		.157 .147	.059 .049	.0450 .0430
# 5	.026	.023	.075	.071				.175 .165	.066 .056	.0500 .0480
# 6	.029	.026	.082	.078	.030	.016		.193 .183	.072 .062	.0555 .0535
# 8	.034	.031	.096	.092				.228 .218	.085 .075	.0660 .0640
#10	.039	.036	.111	.107	.040	.023		.263 .253	.098 .088	.0760 .0740
1/4	.051	.048	.147	.143			.016	.345 .335	.128 .118	.0990 .0970
5/16	.064	.061	.184	.180	.052	.031		.382 .372	.134 .124	.0985 .0965
3/8	.076	.073	.220	.216				.457 .447	.161 .151	.1180 .1160
7/16	.089	.086	.256	.252	.072	.048		.498 .488	.169 .159	.1205 .1185
1/2	.102	.098	.292	.288	.089	.062		.568 .558	.193 .183	.1375 .1355
9/16	.115	.111	.328	.324	.108	.078	.024	.638 .628	.217 .207	.1545 .1525
5/8	.127	.123	.365	.361				.708 .698	.241 .231	.1710 .1690

NOTES:

1. All dimensions are in inches.
2. Material: Alloy steel in accordance with AMS 6485, (UNS T20811) or AMS 6487, (UNS T20811).
3. Heat treatment: 270,000 PSI minimum tensile strength 52-56HRC.
4. Finish: None.
5. Recesses to be gaged per MS33781 except gage penetration is limited to this drawing.

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9. Revision letters are not used to denote changes due to the extensiveness of the changes.

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Navy - AS
Air Force - 99

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Air Force - 80
DLA - IS

User Activities:

Preparing Activity:
Air Force - 84

Agent:
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